



Looping Statements: -

In Our program we need to execute block of code number of known or unknown times. To implement such scenario looping statements are used by programmers.

Below are the looping statements available in JavaScript.

1. for loop.
2. for of loop.
3. for in loop.
4. while loop.
5. do while loop.

1. for loop: -

JavaScript for loop is used to execute block of code known times.

JavaScript for loop accepts three parameter arguments.

1. First parameter defines the variable initialization.
2. Second parameter is use to define condition of for loop.
3. Third parameter is use to increase or decrease the value of variable by one, after each loop executes the block of code.



```
// * pattern program:-  
  
//  initialization ;  condition ; increment  
for(  let i=1;          i<=10;      i++)  
{  
    for(let j=1;j<=i;j++)  
    {  
        document.write("*")  
    }  
    document.write("<br>")  
}  
  
// Array iterate using array  
const days=['SUN','MUN','TUE','WED','THU','FRI','SAT'];  
  
for(let i=0;i<days.length;i++)  
{  
    document.write(days[i]);  
    document.write('<br>')  
}
```

2. for of loop: -

JavaScript for of loop is special type of loop, that travers over array element.

```
// for of loop:-  
for(let day of days)  
{  
    document.write(`<h1>${day}</h1>`)  
}
```



3. for in loop: -

JavaScript for in loop is special loop that, travers over object properties.

Sometimes, we don't know which properties JavaScript object contains, in order to know object's properties we use for in loop.

```
let employee={
  empId:101,
  empName:'Gajanan',
  empDesignation:'Developer',
  empSalary:45000
}
for(let prop in employee)
{
  document.write(`<h1>${prop} :- ${employee[prop]}</h1>`)
}
```

Output: -

empId :- 101
empName :- Gajanan
empDesignation :- Developer
empSalary :- 45000



4. while loop: -

The while loop loops through a block of code as long as a specified condition is true.

5. do while: -

The do while loop is a variant of the while loop. This loop will execute the block of code once, before checking if the condition is true, then it will repeat the loop as long as the condition is true.

Spread Operator: -

In JavaScript spread operators are used to copy one object property into the other object.

The symbol of spread operator is `...`.

The spread operator is immediately followed by the object reference which is supposed to be copied.

Example: - we have u1 and u2 object, and the u1 needs to be copied in u2

```
const u1={
  id:101,
  username:"samrth",
  password:"samrth@123"
}
console.log(u1); //{ id:101,username:"samrth",password:"samrth@123"}
```



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```
const u2={
  ...u1, // this will copy all properties and their values from u1 to u2
  email:"samrthk123@gmail.com"
}
console.log(u2); // { id:101,username:"samrth",password:"samrth@123",
                  // email:"samrthk123@gmail.com" }
```

Note: - Using spread operator we can copy an array into another array to.

Example: - We have nos1 array and nos2 array, we need to copy the elements of nos1 array into nos2 array.

```
const nos1=[10,20,30,40,50];
console.log(nos1);           // [10, 20, 30, 40, 50]
const nos2=[...nos1,60,70,80,90,100];

console.log(nos2);           // [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
```

Rest Parameter: -

The rest parameter syntax allows a function to accept infinite number of arguments as an array.

```
function add(...params)
{
  let result=0;
  for(let no of params)
  {
    result +=no;
  }
  console.log(result)
```



```
}  
  
add(10,20,30,40,50,60)
```

De-structuring: -

The de-structuring assignment syntax unpack object/array values into variables.

In case of array de-structuring sequence of array index is followed.

In case of object de-structuring sequence of object properties can be avoid.

```
const student={  
  rollno:1,  
  firstName:'aaa',  
  lastName:'zzzz',  
  batchNo:'FDJ-176',  
  classroom:'ground class',  
  course:'Java Fullday'  
}  
  
console.log(student.rollno);  
console.log(student.batchNo);  
  
const {rollno,batchNo,classroom}=student  
  
document.write(`<h1>${rollno}</h1>`)  
document.write(`<h1>${batchNo}</h1>`)  
document.write(`<h1>${classroom}</h1>`)  
  
// 0 1 2 3 4 5 6
```



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```
const days=['SUN','MUN','TUE','WED','THU','FRI','SAT'];

// let i=days[0];
// let j=days[1];

const [i,j,,,k]=days

document.write(`<h1>${i}</h1>`);
document.write(`<h1>${j}</h1>`);
document.write(`<h1>${k}</h1>`);
```



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Complete Java Classes